

Quantum Threats & Post-Quantum Cryptography

What the threat actually is

Shor's algorithm (quantum) can factor integers and solve discrete log in polynomial time → breaks textbook RSA, DH, ECDSA/ECDH – the internet's public-key layers.

Grover's algorithm (quantum) square-roots brute force → 128-bit AES ≈ 64-bit effective for search; solvable with longer keys (AES-256, SHA-384 stay comfortable).

Timeline honesty

- Quantum computers are small; no one breaks RSA-2048 today
- But 'Harvest now, decrypt later' is real: encrypted data recorded today becomes plaintext once long-term-secure quantum arrives
- Transition windows: years of migration; a patient adversary waits

The NIST post-quantum winners (announced 2024)

Algorithm	Purpose	Notes
ML-KEM (Kyber)	Key encapsulation (replaces DH/ECDHE)	Lattice-based
ML-DSA (Dilithium)	Digital signatures	Lattice-based
SLH-DSA (SPHINCS+): stateless hash signatures	Backup, hash-based	Slower, huge signatures, ultra-conservative

Plus Falcon/KNIGHT variants for special use.

What changes for defenders

Key exchange: hybrid TLS (X25519 + ML-KEM) – so today's PFS works and quantum-proofing lands atomically

Today's public key ops: RSA/ECDH→ ML-KEM/ML-DSA in standards (RFC 8784, TLS 1.3 5093-ish hybrids)

Post-quantum keys: bigger; signature sizes differ (use-case aware)

Practical prep (even today)

1. Inventory crypto use: TLS, code signing, email (S/MIME & OpenPGP), VPN, HSM
2. Prefer hybrid stacks now (X25519 + ML-KEM combo in your VPN/TLS)
3. Validate your CA/provid-w will rotate to PQC
4. Long-term data: apply 'harvest-now' lens (PQC-encrypt archival)
5. Watch NIST IR / vendor mappings as they land

The philosophical point

Crypto is clock-driven: you plan for the break *before* the computer exists, because data lasts longer than crypto reputations.

Post-quantum is not 'next decade' – it's 'this decade's migration plan'. The professional stack: hybrid now, inventory crypto, prioritize long-lived secrets.

[← Hash Collisions, Length Exte...](#)

Module complete ✓